

TarMAC: Targeted Multi-Agent Communication

Condensed Paper Report - Conclusions, Gaps & Implementation Considerations

Das et al., ICML 2019 | arXiv:1810.11187

Scope of this note

This is deliberately not a full project roadmap. It records what this paper establishes, what it leaves unresolved, and which implementation choices / experiments are justified to consider because of TarMAC.

1 What the paper is mainly about

TarMAC extends emergent multi-agent communication beyond broadcast messaging. The central question is not only what agents should communicate, but whom a message should be relevant to. Agents learn targeted continuous communication solely from downstream task reward, without labels for message meaning or intended recipients.

Core mechanism

- Each agent is a recurrent policy (1-layer GRU) that receives its local observation plus the aggregated communication from the previous step and outputs both an environment action and a continuous message.
- A message is split into a key/signature k (describing properties of intended recipients) and a value v (the actual message content). Each receiver produces a query q from its hidden state.
- Receiver-specific attention is computed from query-key similarity; the received communication is an attention-weighted sum of message values. Targeting is implicit: the system learns recipient properties, not explicit agent IDs.
- Self-attention is allowed and was found useful, e.g., an agent already at its goal can preserve its own state while other agents attend to it.
- TarMAC also supports multiple communication rounds between environment actions, allowing back-and-forth collaborative reasoning before acting.
- Training uses a centralized critic and decentralized actor execution. In the paper experiments, policy parameters are shared across symmetric agents.
- The communication path is fully differentiable; continuous messages are learned end-to-end by backpropagation.

Useful conceptual split

TarMAC mainly addresses WHO should be listened to (receiver-dependent targeting). It complements methods such as IC3Net that address WHEN an agent should communicate at all.

2 Experimental evidence and conclusions

The paper evaluates four environments. The important evidence is the pattern across the ablations, not just the best final number.

2.1 SHAPES - targeted communication matters more as information needs diverge

Setting	No comm	Broadcast / no attention	TarMAC
30x30, same goal	95.3 +/- 2.8%	99.7 +/- 0.8%	99.8 +/- 0.9%
50x50, same goal	83.6 +/- 3.3%	89.5 +/- 1.4%	89.5 +/- 1.7%

Setting	No comm	Broadcast / no attention	TarMAC
50x50, different goals	69.1 +/- 4.6%	82.4 +/- 2.1%	85.8 +/- 2.5%

- Communication helps under partial observability. The additional benefit of attention is small when agents need similar information, but increases in the harder different-goal setting.
- Attention visualizations are qualitatively intuitive: agents shift attention toward agents observing a relevant target and later self-attend when their own goal is reached.

2.2 Traffic Junction - targeting + multi-round communication is the strongest result

Method	Easy	Hard
No communication	84.9 +/- 4.3%	74.1 +/- 3.9%
CommNet	99.7 +/- 0.1%	78.9 +/- 3.4%
TarMAC, 1 round	99.9 +/- 0.1%	84.6 +/- 3.2%
TarMAC, 2 rounds	99.9 +/- 0.1%	97.1 +/- 1.6%

- On the hard task, 2-round TarMAC reaches 97.1% success, about 18 percentage points above CommNet; the paper reports no further gains beyond two rounds in this environment.
- Increasing message dimensionality from 1 to 64 helps much less than adding a second communication round. Even a 1-D continuous scalar performed nearly as well as 64-D in this task.
- Learned attention focuses on traffic-sensitive locations. The number of agents receiving substantial attention rises with the number of active cars (Spearman correlation 0.49; 0.53 after a 3-step shift), suggesting adaptation to dynamic team size.

2.3 House3D - communication improves both success and efficiency

Method	Success	Avg. steps
No communication	62.1 +/- 5.3%	186.5
Broadcast / no attention	64.3 +/- 2.3%	101.3
TarMAC	68.9 +/- 1.1%	82.5

- TarMAC yields the best success and substantially shorter trajectories in a first-person 3D navigation task. This shows that communication may improve efficiency even when success-rate gains are modest.
- The paper uses shaped progress reward in House3D. “Task reward only” means communication itself is not supervised; it does not imply every task uses sparse reward.

2.4 Mixed / competitive Predator-Prey - targeting needs a communication gate

Method	3 agents	5 agents	10 agents
CommNet	9.1	13.1	76.5
IC3Net	8.9	13.0	52.4
IC3Net + TarMAC	8.31	12.74	41.67
IC3Net + TarMAC, 2 rounds	7.24	-	35.57

- Soft attention always leaves low but non-zero “leakage”, which is problematic when an agent has an incentive not to reveal information. The authors therefore combine TarMAC with IC3Net hard communication gating.

- The combined system learns both WHEN to communicate and WHOM to target, converges faster, and uses fewer steps at convergence. These results use 3 seeds (other experiments generally use 5).

3 What the paper supports as conclusions

- 1. Communication under partial observability:** Learned inter-agent messages can materially improve cooperative performance when agents hold complementary local information.
- 2. Targeting is conditionally useful:** Receiver-dependent attention is most valuable when different agents need different information; on simple homogeneous tasks, broadcast communication can be almost as good.
- 3. Interaction structure can matter more than raw vector width:** A second communication round produced a much larger gain than increasing continuous message dimension in Traffic Junction.
- 4. Targeting can adapt with population changes:** Attention patterns changed with the number of active agents without explicit supervision.
- 5. Learned communication can be inspectable:** Attention matrices give a useful view of who is listening to whom, although they do not by themselves prove the semantic meaning of a message.
- 6. WHEN and WHOM are complementary:** A hard communication gate and targeted attention solve different problems and can be combined.

Critical interpretation of the “1-D message” result

A single continuous scalar is not a one-bit channel. A real-valued scalar can encode many distinguishable states, so this result does not demonstrate true low-bandwidth communication. Message dimensionality and network bit-rate must be treated separately.

4 Important limitations and gaps

The following gaps are either explicit limitations acknowledged by the paper or important consequences of its experimental assumptions. These are not claims that TarMAC is “wrong”; they define what its results do not establish.

- 1. Targeting is semantic, not necessarily network-level unicast.** Every sender broadcasts a key/value representation and receivers perform attention. The method decides whose information matters, but does not by itself prove reduced transmitted bytes or sparse radio traffic.
- 2. Communication is effectively ideal.** The experiments do not model latency, packet loss, corruption, interference, congestion, explicit communication energy/cost, or realistic wireless scheduling.
- 3. Communication range is mostly unconstrained.** For example, cars have very local sensing but are free to communicate with all cars. Results therefore do not establish robustness under range-limited or intermittent links.
- 4. Softmax attention cannot naturally mean “nobody”.** Attention weights distribute mass across messages; in mixed/competitive tasks this creates information leakage. A separate gate/null-message mechanism is needed when silence matters.
- 5. Continuous differentiable channels are easier than deployed digital links.** Backpropagation through continuous messages gives strong credit assignment but does not address quantization, discrete-symbol learning, packet encoding, or deployment bit budgets.
- 6. The paper uses symmetric shared policies.** Its experiments share actor parameters across agents. It does not directly evaluate strongly heterogeneous roles with different capabilities/action spaces.
- 7. Multi-round communication is treated as free pre-action reasoning.** Extra rounds do not incur an explicit environment-time or network-latency cost, so “2 rounds > 1 round” may change when each round is expensive.
- 8. Generalization claims are limited.** Dynamic attention is demonstrated as the active population changes, but this is not a proof of robust zero-shot scaling to arbitrary unseen team sizes/topologies.
- 9. Interpretability is suggestive, not causal decoding.** Attention plots are intuitive, but claims such as a key “encoding color” are interpretations. Message meaning would need intervention/ablation tests to establish causality.
- 10. Reward design still matters.** House3D uses shaped progress reward. Unsupervised communication semantics do not remove the need to design a learnable task reward.

Paper-stated future work

- Force or ground continuous communication into discrete symbols.
- Benchmark more challenging 3D navigation where decentralized targeted communication may matter more.
- Combine TarMAC with spatial memory, planning networks, and related advances.

5 Implementation considerations contributed by TarMAC

These are the concrete implementation ideas this paper justifies considering for our system. They are options/ablation factors, not a final architecture decision.

Consideration	What to implement / log	Why TarMAC makes it relevant
Targeted communication module	Key / query / value sender-receiver attention; log the full attention matrix $A_{ij}(t)$.	Directly tests the paper's main contribution.
Recurrent actor state	GRU/LSTM-style memory over local observations + received communication.	Appropriate for partial observability and message history.
CTDE compatibility	Use decentralized actors with a centralized critic during training.	TarMAC is compatible with this structure; its exact 2019 actor-critic need not be copied.
Self-attention	Allow an agent to include its own value in the attention pool, or ablate it.	Paper reports self-attention as beneficial in some coordination states.
Multi-round communication	Compare 1 vs 2 communication rounds before action.	Strongest TarMAC ablation; >2 rounds need not be default.
Message size	Sweep a small set of value dimensions (e.g., low / medium / high).	Tests whether bandwidth proxy matters relative to interaction rounds.
Communication gate	Optional hard/soft gate before targeting.	Needed if silence, cost, privacy, or competitive incentives matter.
Role conditioning	If agents are heterogeneous, compare shared policy + role embedding vs role-specific heads/policies.	TarMAC shared symmetric policies do not settle this for specialized roles.

6 Experiments / comparisons this paper tells us to consider

These are the paper-derived comparisons worth preserving in the experimental plan. They isolate the effect of communication and targeting without making TarMAC responsible for the entire project design.

Comparison	Question it answers
No communication vs communication	Does inter-agent information sharing help at all in our task?
Broadcast / mean-pooled vs targeted attention	Does learned recipient relevance add value beyond simply sharing messages?
1 round vs 2 rounds	Does iterative collaborative reasoning improve coordination?
Message dimension sweep	Is performance sensitive to continuous message capacity, and how does that compare with extra rounds?
Homogeneous vs heterogeneous information needs	Does targeting become more useful when different roles/goals require different information?

Comparison	Question it answers
Targeting alone vs targeting + gate	Does explicit silence improve efficiency/robustness when not every agent should speak?
Ideal channel vs delay/loss/range/bandwidth constraints	Do TarMAC conclusions survive when communication has physical/network cost?
Continuous vs quantized/discrete messages	Does performance remain when “message dimension” is converted into an actual bit-budget constraint?
Attention visualization + intervention	Do learned attention patterns correspond causally to useful information flow?
Team-size / topology variation	How well does learned targeting adapt when the number or connectivity of agents changes?

Metrics implied by the paper

- Task success / completion rate and episode return.
- Completion steps / time-to-goal, not only final success.
- Training sample efficiency / convergence speed.
- Attention sparsity / recipient counts / sender-receiver patterns.
- Under realistic channel tests: communication volume, delay sensitivity, packet-loss sensitivity and true bit cost.

7 What NOT to conclude from this paper

- TarMAC is not proven to beat broadcast communication on every task; its advantage grows when recipient relevance is heterogeneous.
- Two communication rounds are not universally optimal; the paper only reports that >2 gave no extra gain in its Traffic Junction test.
- A 1-D continuous message is not evidence that one bit is sufficient.
- Attention targeting is not automatically equivalent to reducing network transmissions.
- The paper does not decide our RL optimizer/backbone or justify copying its RMSProp / hidden-size hyperparameters verbatim.
- The paper does not determine the whole project direction; it supplies communication mechanisms, baselines and gaps that should be tested in our own environment.

8 Paper-specific takeaway

Bottom line

TarMAC provides strong evidence that communication should be treated as more than a binary “on/off” feature. In tasks where agents need different information, learned sender-receiver targeting can outperform broadcast communication, and iterative communication rounds can matter more than simply increasing continuous message width. For our work, the paper primarily contributes a targeted-attention implementation candidate, the no-comm / broadcast / targeted baselines, the 1-vs-2-round and message-size ablations, and a set of unresolved realism questions around gating, latency, loss, range, quantization and actual transmission cost.

Reference: A. Das et al., “TarMAC: Targeted Multi-Agent Communication,” *Proceedings of the 36th International Conference on Machine Learning (ICML)*, PMLR 97, 2019. *arXiv:1810.11187*. Paper version consulted: *arXiv v2* (22 Feb 2020).